Restaurant ERP – Software Requirements Specification (SRS)

Version 1.0 · Scalable platform for small, medium and large restaurant operations · Multi-site, workspace-based

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1. Introduction

1.1 Purpose

This document specifies the requirements for **RestoERP**, a cloud-first, modular Enterprise Resource Planning platform for restaurants. It must serve a single-outlet café, a growing multi-branch chain, and a large franchise or central-kitchen group using the same core product, activated by tier and modules.

1.2 Scope

RestoERP covers the full restaurant value chain: front-of-house sales, kitchen operations, inventory and recipe costing, procurement, central production, finance, HR, customer engagement, analytics and head-office control. Each physical site receives its own **Site Workspace**, while owners and head office see consolidated **HQ Workspaces**.

1.3 Definitions

Term	Meaning
Tenant	One customer company (owner/group)
Brand	A restaurant concept under a tenant
Site	A physical outlet, kitchen, warehouse or kiosk
Workspace	Role-based, site-scoped working environment with its own dashboards, tasks and permissions
POS	Point of Sale
KDS	Kitchen Display System
BOM / Recipe	Ingredient list with quantities and yield

Term	Meaning
CK	Central Kitchen
PAR level	Target stock level per item per site
COGS	Cost of goods sold

1.4 Intended audience

Product owners, business analysts, architects, developers, QA, implementation consultants and restaurant management.

1.5 Product vision

One system that is simple enough for a 10-seat shop to start in a day, and deep enough for a 500-site chain to control cost, quality and growth.

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2. Overall Description

2.1 Product tiers

Capability	Starter (1–2 sites)	Growth (3–25 sites)	Enterprise (25+ sites, franchise, CK)
POS, tables, KDS, basic reports	✔	✔	✔
Inventory and recipes	Basic	Full	Full + forecasting
Procurement	Simple PO	Approval flows	Contracts, tenders, 3-way match
Multi-site HQ	–	✔	✔ + franchise royalty engine
Central kitchen	–	Optional	✔
HR, payroll, scheduling	Basic roster	Full	Full + labor optimization
Accounting	Export to external	Built-in GL	Multi-entity consolidation

Capability	Starter (1–2 sites)	Growth (3–25 sites)	Enterprise (25+ sites, franchise, CK)
BI and AI forecasting	Standard reports	Dashboards	Custom BI, ML forecasting
API / integrations	Limited	Standard	Full, webhooks, SSO

Modules are feature-flagged per tenant so customers pay for and see only what they need. Upgrading never requires data migration.

2.2 User classes

- **Owner / Executive** – cross-site performance, profit, cash.
- **Regional / Area Manager** – compares and controls a cluster of sites.
- **Site (General) Manager** – runs one outlet daily.
- **Cashier / Waiter** – orders, payments, tables.
- **Kitchen Chef / Line Cook** – tickets, prep, waste.
- **Storekeeper** – receiving, issuing, stock counts.
- **Purchasing Officer** – suppliers, orders, price control.
- **Accountant / Finance** – ledgers, tax, reconciliation.
- **HR Officer** – staff, attendance, payroll.
- **Marketing / CRM Officer** – loyalty, promotions.
- **Franchisee** – own-site view only.
- **Auditor (read-only)** and **System Administrator**.
- **Customer** – ordering, reservations, loyalty (web/app).
- **Supplier** – portal for orders and invoices.

2.3 Operating environment

- Cloud SaaS (multi-tenant) with optional private/on-premise deployment for enterprise.
- Clients: Web (responsive PWA), Android/iOS tablet POS, handheld waiter app, KDS screens, customer web/app, supplier portal.
- Hardware: receipt/kitchen printers (ESC/POS, network/USB/Bluetooth), cash drawers, barcode scanners, scales, card/mobile-money terminals, fiscal devices, label printers.

2.4 Design constraints

- Must keep trading when the internet fails (offline-first POS).
- Must support multi-currency, multi-language (incl. RTL), multi-tax, local fiscal regulations.
- Must run acceptably on low-cost tablets and low bandwidth.

2.5 Assumptions

Sites have at least intermittent connectivity; each site has a designated manager; master data (menu, items, suppliers) can be shared across sites but overridden locally.

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3. Organization and Workspace Model (core of the design)

3.1 Hierarchy

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Tenant (Company Group)
 └─ Legal Entity (tax registration, currency)
     └─ Brand
         └─ Region / Area
             └─ Site (Restaurant, Kiosk, Cloud Kitchen, Central Kitchen,
Warehouse)
                 └─ Station / Terminal (POS 1, Bar, Grill, Pass, Store)
```

Every transaction carries tenant, entity, brand, region, site, station, user and timestamp.

3.2 Workspace types

1. **Group HQ Workspace** – owners/executives: consolidated KPIs, P&L by site, cash position, alerts, approvals.
2. **Regional Workspace** – benchmarking, audit checklists, stock transfers, regional promotions.
3. **Site Workspace** – one per outlet; the daily operating console described in section 5.
4. **Central Kitchen / Warehouse Workspace** – production plans, internal orders, dispatch.
5. **Franchisee Workspace** – isolated view of own sites plus royalty statements.
6. **Procurement and Finance Workspaces** – cross-site functional views.

3.3 Workspace rules

- **WS-01** Creating a site automatically provisions a Site Workspace from a template (roles, dashboards, printers, tax, menu, floor plan).
- **WS-02** Each workspace has its own dashboards, task inbox, notifications, calendar, and daily checklists.
- **WS-03** Data isolation: a user sees only workspaces and data assigned to them; cross-site visibility needs explicit grants.
- **WS-04** Inheritance: menu, prices, recipes, taxes, policies are defined at brand/group level and may be overridden per site with audit trail.
- **WS-05** Site templates and cloning allow opening a new outlet in under one day.
- **WS-06** Site lifecycle states: Draft → Pilot → Active → Temporarily Closed → Archived, with data retention.
- **WS-07** Users can belong to multiple workspaces with different roles (e.g., manager of Site A, viewer at Site B) and switch via a workspace switcher.
- **WS-08** Day-end closing per site locks the business day, reconciles cash and stock, and pushes results to HQ.

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4. Functional Requirements (by module)

Priority: **M** = Must, **S** = Should, **C** = Could.

4.1 Point of Sale (POS)

ID	Requirement	Pri
POS-01	Dine-in, takeaway, delivery, drive-through, bar tab, kiosk and self-order modes	M
POS-02	Table map with drag-and-drop floor plans, merge/split/transfer tables and checks	M
POS-03	Modifiers, combos, courses, allergen flags, special notes, seat numbers	M
POS-04	Split bills (by item, seat, amount, percentage); partial and mixed payments	M

ID	Requirement	Pri
POS-05	Payments: cash, card, mobile money, QR, vouchers, gift cards, loyalty points, house accounts, tips	M
POS-06	Discounts, promotions, happy hour, coupons with role-based limits and approval	M
POS-07	Void/refund/comp with reason codes, manager PIN, full audit	M
POS-08	Offline mode: continue selling, printing and payments (cash) for at least 72 hours; automatic conflict-free sync	M
POS-09	Fiscal receipt integration (e.g., EFD/VFD/e-invoicing per country), tax-inclusive and exclusive pricing	M
POS-10	Shift open/close, cash drops, blind cash count, over/short report	M
POS-11	Price scheduling by time, channel, site, or customer group	S
POS-12	Waiter handheld ordering with fire-to-kitchen	M
POS-13	Customer-facing display and QR self-ordering at table	S
POS-14	Voice/AI upsell suggestions at order entry	C

4.2 Kitchen Operations and KDS

ID	Requirement	Pri
KDS-01	Route items to stations (grill, fry, cold, bar, pastry) by rules	M
KDS-02	Ticket timers, color-coded urgency, bump, recall, expo/pass screen	M
KDS-03	Course firing and hold	M
KDS-04	Printer fallback if a KDS screen fails	M

ID	Requirement	Pri
KDS-05	Prep lists computed from forecast and open orders	S
KDS-06	86-ing (out-of-stock) propagated live to POS, online and kiosks	M
KDS-07	Kitchen performance analytics (ticket time per station and item)	S

4.3 Reservations, Waitlist and Events

- **RES-01 (M)** Online and phone reservations with table assignment, capacity rules, turn-time estimates.
- **RES-02 (M)** Waitlist with SMS/WhatsApp notification.
- **RES-03 (S)** Deposits and no-show fees.
- **RES-04 (S)** Banquet/event management: quotes, packages, deposits, BEO sheets, headcount-based production.

4.4 Online Ordering, Delivery and Aggregators

- **ONL-01 (M)** Branded website/app ordering with live menu, availability and site selection.
- **ONL-02 (M)** Aggregator integration (marketplaces, local delivery apps): orders injected to POS/KDS, menu and availability sync, commission tracking.
- **ONL-03 (M)** Delivery management: zones, fees, driver assignment, live tracking, proof of delivery.
- **ONL-04 (S)** Cloud/ghost kitchen support: virtual brands sharing one kitchen.

4.5 Menu, Pricing and Recipe Management

- **MNU-01 (M)** Central menu with categories, items, variants, modifiers, combos, allergens, nutrition, images, multi-language names.
- **MNU-02 (M)** Recipe/BOM with sub-recipes, yields, wastage %, unit conversions.
- **MNU-03 (M)** Real-time recipe costing from latest purchase cost; margin and food-cost % per item.
- **MNU-04 (S)** Menu engineering matrix (stars, plowhorses, puzzles, dogs).
- **MNU-05 (S)** Menu versioning, scheduled publishing, approval workflow.

- **MNU-06 (M)** Price lists per site, channel, and customer group.

4.6 Inventory and Warehouse

ID	Requirement	Pri
INV-01	Item master: stock, non-stock, prep, packaging, assets; multiple UoM; barcodes/SKUs	M
INV-02	Multi-location stock (store, kitchen, bar, freezer, CK, warehouse)	M
INV-03	Receiving against PO with quality check, temperature log, partial delivery	M
INV-04	Automatic depletion by recipe on sale (theoretical usage)	M
INV-05	Stock count: full, cycle, blind; variance (actual vs theoretical) with approval	M
INV-06	Inter-site and inter-location transfers with in-transit tracking	M
INV-07	Batch/lot and expiry tracking, FEFO issuing, expiry alerts	M
INV-08	Waste and spoilage logging with reason, photo, cost	M
INV-09	PAR levels, reorder points, auto-suggested requisitions	M
INV-10	Costing methods: weighted average, FIFO, last cost, standard cost	M
INV-11	Mobile stock counting with barcode scanning and scale integration	S
INV-12	Demand forecasting using sales history, weather, events, seasonality	S

4.7 Procurement and Supplier Management

- **PRC-01 (M)** Supplier master with catalogs, contracts, lead times, price history, ratings.

- **PRC-02 (M)** Requisition → RFQ → PO → Goods Receipt → Invoice (3-way match).
- **PRC-03 (M)** Configurable approval chains by amount, category and site.
- **PRC-04 (S)** Supplier portal: view POs, confirm, upload invoices, track payments.
- **PRC-05 (S)** Price variance alerts and best-price recommendation.
- **PRC-06 (C)** EDI and e-ordering to major suppliers.

4.8 Central Kitchen and Production

- **CK-01 (M)** Internal orders from sites to CK; consolidated production plan.
- **CK-02 (M)** Production orders with BOM, yield, batch numbers, labor and overhead costing.
- **CK-03 (M)** Dispatch, delivery notes, route planning, receiving confirmation at sites.
- **CK-04 (S)** HACCP records, label printing with date, batch and allergens.

4.9 Finance and Accounting

ID	Requirement	Pri
FIN-01	General ledger, chart of accounts templates per country, cost centers by site/brand	M
FIN-02	Accounts payable and receivable, aging, payment runs	M
FIN-03	Automatic journal posting from sales, purchases, inventory, payroll	M
FIN-04	Daily sales and bank/mobile-money reconciliation, payment-provider fee handling	M
FIN-05	Tax engine: VAT/GST/sales tax, service charge, withholding; compliance reports	M
FIN-06	Budgeting and forecasting per site; actual vs budget	S
FIN-07	Petty cash and expense management with receipt capture	M
FIN-08	Fixed assets and depreciation (equipment)	S

ID	Requirement	Pri
FIN-09	Multi-currency, multi-entity consolidation and intercompany	S
FIN-10	Export/integration with external accounting systems (for Starter tier)	M
FIN-11	Franchise royalty, marketing fund and fee calculation and invoicing	S

4.10 Human Resources, Scheduling and Payroll

- **HR-01 (M)** Employee records, contracts, documents, certifications (food-handler, medical) with expiry alerts.
- **HR-02 (M)** Roster/shift planning with demand-based labor targets, availability, swaps, labor cost % live.
- **HR-03 (M)** Time and attendance: PIN, biometric, geofenced mobile clock-in; break rules; overtime.
- **HR-04 (M)** Payroll: salaries, hourly, tips distribution, commissions, allowances, statutory deductions per country, payslips.
- **HR-05 (S)** Onboarding, training tracking, performance reviews, staff meals.
- **HR-06 (S)** Leave management and approvals.

4.11 Customer Relationship and Marketing

- **CRM-01 (M)** Customer profiles, visit history, preferences, allergies.
- **CRM-02 (S)** Loyalty: points, tiers, stamps, wallet, referral.
- **CRM-03 (S)** Campaigns by SMS, email, WhatsApp, push; segmentation; promo code tracking and ROI.
- **CRM-04 (S)** Feedback and reviews capture (QR, post-visit survey), alerts on low scores.
- **CRM-05 (C)** Gift cards and e-vouchers.

4.12 Reporting, Analytics and BI

- **RPT-01 (M)** Standard reports: sales by item/category/hour/channel/server, voids, discounts, payments, tax, shift, COGS, waste, variance, labor, supplier spend.

- **RPT-02 (M)** Role dashboards in every workspace (see section 5) with drill-down from group to site to order.
- **RPT-03 (M)** Scheduled reports by email/WhatsApp; export to CSV/Excel/PDF.
- **RPT-04 (S)** Site benchmarking and ranking by KPI.
- **RPT-05 (S)** Self-service report builder and data warehouse export.
- **RPT-06 (C)** AI insights: anomaly/fraud detection, forecast, "why did sales drop" narrative.

4.13 Quality, Safety and Compliance

- **QSC-01 (M)** Digital checklists (opening, closing, cleaning, safety) with photos and manager sign-off.
- **QSC-02 (M)** Temperature and HACCP logs (manual or IoT sensor).
- **QSC-03 (S)** Audit scheduler with scoring and corrective-action tracking.
- **QSC-04 (S)** Allergen and traceability reports (forward/backward by batch).

4.14 Asset and Maintenance

- **AST-01 (S)** Equipment register, maintenance schedules, breakdown tickets, vendor contacts, warranty.
- **AST-02 (C)** IoT monitoring for fridges and ovens.

4.15 Administration and Configuration

- **ADM-01 (M)** Tenant, entity, brand, region, site, station setup wizards.
- **ADM-02 (M)** Role-based access control with granular permissions, custom roles, field-level restrictions.
- **ADM-03 (M)** Approval workflow designer (discounts, POs, refunds, stock adjustments, expenses).
- **ADM-04 (M)** Audit log of every create/update/delete/approve action, tamper-evident.
- **ADM-05 (M)** Data import tools (menu, items, suppliers, opening stock, staff) with validation.
- **ADM-06 (S)** Notification center with rules (email, SMS, push, WhatsApp).
- **ADM-07 (S)** Localization packs: language, currency, tax, fiscal, payroll rules.

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5. Site Workspace Design (per outlet)

Each site gets a well-organized workspace with a left navigation, a role-aware home dashboard, a task inbox and a day timeline (Open → Service → Close).

5.1 Common layout

- **Top bar:** site name, business date, shift status, connectivity indicator, workspace switcher, notifications.
- **Home:** today's sales vs target, covers, average check, live orders, labor %, stock alerts, pending approvals.
- **Task inbox:** checklists, approvals, receiving, counts due, expiring items, maintenance tickets.
- **Quick actions:** new order, receive delivery, log waste, start count, clock in.

5.2 Role-based areas inside the Site Workspace

Area	Primary users	Contents
Front of House	Cashier, waiter, host	POS, floor plan, reservations, waitlist, online order queue
Kitchen & Bar	Chef, cooks, bartender	KDS, prep lists, recipe cards, 86 list, waste log, temperature logs
Store & Inventory	Storekeeper	Receiving, issue, transfers, counts, expiry, requisitions
Purchasing (site)	Manager, storekeeper	Site requisitions, POs, supplier deliveries, invoice matching
Cash & Finance (site)	Manager, cashier	Shift reports, cash drops, petty cash, daily close, deposits
People	Manager, HR	Roster, attendance, leave, training, certifications
Customers	Manager, marketing	Feedback inbox, loyalty lookups, complaints, promotions active
Quality & Maintenance	Manager, chef	Checklists, audits, equipment tickets

Area	Primary users	Contents
Reports	Manager	Site dashboards and standard reports
Settings	Manager (limited)	Printers, stations, floor plan, local price overrides, staff PINs

5.3 Day lifecycle automation

1. **Opening:** checklist, float count, stock-critical alerts, roster check, prep list.
2. **Service:** live monitoring of tickets, voids, stock-outs, labor vs sales.
3. **Closing:** cash count, checklist, waste entries, end-of-day stock snapshot (key items), auto-close of tabs, Z-report.
4. **Post-close:** automatic journals to finance, HQ dashboard refresh, exception report to area manager.

5.4 Workspace sizing

- **Small site:** simplified single-screen workspace; modules hidden unless enabled.
- **Large site:** department sub-workspaces (Bar, Banquet, Bakery, Delivery) with separate stations and budgets.

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6. Integration Requirements

- **INT-01** Public REST and GraphQL APIs, webhooks, API keys/OAuth 2.0 with rate limits.
- **INT-02** Payment gateways, card terminals, local mobile-money providers, QR payments.
- **INT-03** Delivery marketplaces, reservation platforms, Google Business, social ordering.
- **INT-04** Accounting (e.g., QuickBooks, Xero, Sage), payroll providers, banks (statement import).
- **INT-05** Fiscal/tax authorities' e-invoicing and device APIs.
- **INT-06** Messaging: SMS gateways, WhatsApp Business API, email.
- **INT-07** Hardware via a local agent: printers, drawers, scales, scanners, biometric devices.
- **INT-08** SSO (SAML/OIDC), directory sync for enterprises.
- **INT-09** Data export to data warehouses and BI tools.

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7. Non-Functional Requirements

7.1 Performance and scalability

- **NFR-01** POS action response under 300 ms locally; order send to KDS under 1 s on LAN.
- **NFR-02** Support 10,000+ sites and 50,000 concurrent users per deployment with horizontal scaling.
- **NFR-03** Reports on one site for one year in under 5 s; group dashboards under 10 s using pre-aggregated data.

7.2 Availability and reliability

- **NFR-04** 99.9% cloud uptime (99.95% for Enterprise); RPO ⩽ 5 min, RTO ⩽ 1 h.
- **NFR-05** Offline-first local store at each site with automatic, idempotent sync and conflict resolution rules.
- **NFR-06** Optional on-site server node (site hub) so devices keep working together when internet is down.

7.3 Security and privacy

- **NFR-07** TLS 1.2+ in transit; AES-256 at rest; secrets in managed vault.
- **NFR-08** MFA for managers and above; PIN/biometric for floor staff; session timeouts.
- **NFR-09** PCI DSS scope reduction via tokenized card handling; never store card numbers.
- **NFR-10** GDPR/local data-protection compliance: consent, export, deletion, data residency options.
- **NFR-11** Tenant isolation at data and encryption-key level; periodic penetration tests.
- **NFR-12** Immutable audit trail and fraud controls (void/discount anomaly alerts).

7.4 Usability

- **NFR-13** New cashier trained in under 2 hours; touch-first UI, large targets, high contrast for bright kitchens.
- **NFR-14** Multi-language UI, RTL support, accessibility (WCAG 2.1 AA for web/customer apps).
- **NFR-15** Guided onboarding wizards, in-app help, and demo data mode.

7.5 Maintainability and extensibility

- **NFR-16** Modular services, feature flags, zero-downtime releases, backward-compatible APIs (versioned).
- **NFR-17** Configurable fields, custom reports, and workflow rules without code changes.
- **NFR-18** Automated test coverage ⩾ 80% on core financial and inventory logic.

7.6 Data and retention

- **NFR-19** Configurable retention per country law (typically 5–10 years for financial data); archival of closed sites.
- **NFR-20** Daily encrypted backups with tested restores.

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8. System Architecture

8.1 Logical architecture

- **Client layer:** PWA/web, native tablet POS, handheld app, KDS, customer app, supplier portal.
- **Edge layer (per site):** optional Site Hub – local database, print server, device agent, sync engine.
- **API gateway:** authentication, tenancy resolution, rate limiting, routing.
- **Domain services:** Sales, Menu, Inventory, Procurement, Production, Finance, HR, CRM, Reporting, Workflow, Notification, Integration.
- **Data layer:** transactional database (PostgreSQL), cache (Redis), event bus (Kafka or equivalent), object storage for files, analytics warehouse (columnar) for BI.
- **Platform services:** identity and RBAC, audit, configuration, localization, job scheduler, search.

8.2 Key architectural decisions

1. **Modular monolith first, services where needed** for small deployments; extract high-load services (Sales, Reporting) as scale grows.
2. **Event-driven:** every sale, receipt, waste or adjustment emits an event; inventory, finance, loyalty and analytics subscribe, ensuring consistency and extensibility.
3. **Multi-tenancy:** shared infrastructure with tenant ID row-level security; dedicated database option for large enterprises.

4. **Offline-first sync:** local event log with unique IDs, replay to cloud, deterministic conflict rules (sales never overwritten; master data last-approved-wins).
5. **CQRS for reporting:** read models and aggregates per site/day so HQ dashboards never slow down POS.
6. **Configuration over customization:** site templates, rules engine and workflow designer.

8.3 Deployment options

- Multi-tenant cloud (default), single-tenant cloud, hybrid (cloud plus site hub), on-premise (rare, enterprise).
- Containerized (Kubernetes), infrastructure as code, multi-region with regional data residency.

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9. Core Data Model (summary)

Organization: Tenant, LegalEntity, Brand, Region, Site, Station, User, Role, Permission, WorkspaceMembership.

Catalog: Item, ItemVariant, Modifier, Combo, Recipe, RecipeLine, PriceList, PriceRule, TaxRule, Allergen.

Sales: Order, OrderLine, Check, Payment, Refund, Shift, CashMovement, Reservation, Table, Customer, LoyaltyAccount.

Inventory: StockItem, Location, StockLedger (immutable movements), Batch, Count, CountLine, Transfer, WasteEntry.

Procurement and Production: Supplier, Requisition, PurchaseOrder, GoodsReceipt, SupplierInvoice, ProductionOrder, DispatchNote.

Finance: Account, JournalEntry, JournalLine, CostCenter, Budget, Tax return, BankStatement, Reconciliation.

People: Employee, Contract, Shift, Attendance, PayrollRun, Payslip, Leave, Certification.

System: AuditLog, WorkflowDefinition, ApprovalRequest, Notification, IntegrationEndpoint, SyncQueue.

Principle: inventory and finance use append-only ledgers; corrections are new reversing entries, never edits.

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10. Key Business Rules

1. A sale posts revenue, tax, payment, theoretical stock depletion and COGS in one atomic event chain.
2. Variance = actual stock minus theoretical stock; variances above threshold require manager reason and approval.
3. Discounts, voids and refunds above role limits require supervisor authorization and are always logged with reason.
4. A closed business day cannot be edited; adjustments are posted in the current open period with reference.
5. Purchase orders above approval limits cannot be sent to suppliers until approved.
6. Items expiring within configured days are blocked from sale/use unless overridden with a reason.
7. Price and menu changes at HQ propagate to sites on a scheduled effective date unless a site override exists.
8. Franchise royalties are computed on net sales per contract and invoiced automatically.

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11. Reports and KPIs

Area	KPIs
Sales	Revenue, covers, average check, sales per labor hour, channel mix, table turn
Cost	Food cost %, beverage cost %, prime cost, waste %, variance %, supplier price index
People	Labor cost %, overtime %, attendance, turnover
Customer	Repeat rate, NPS/rating, loyalty share, delivery time
Control	Voids %, discounts %, refunds, cash over/short, audit scores
Finance	Gross margin, EBITDA per site, cash position, budget variance

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12. Implementation and Rollout Plan

12.1 Phases

Phase	Duration	Deliverables
1. Foundation	3–4 months	Tenancy, org hierarchy, RBAC, workspace engine, menu, POS, KDS, payments, basic reports, offline sync
2. Operations	3 months	Inventory, recipes and costing, procurement, waste, counts, shift/cash management, day-close
3. Business	3 months	Finance/GL, tax and fiscal, HR and rostering, payroll, CRM/loyalty, reservations, online ordering
4. Scale	3–4 months	Multi-site HQ, central kitchen, franchise, benchmarking, approval designer, integrations marketplace
5. Intelligence	ongoing	Forecasting, AI insights, anomaly detection, IoT, advanced BI

12.2 Customer onboarding by size

- **Small:** self-service setup wizard, template menu import, go live in 1 day.
- **Medium:** guided implementation (2–6 weeks): data migration, hardware setup, training per workspace role, pilot site, then roll-out.
- **Large:** program approach (3–9 months): discovery, fit-gap, integrations, pilot cluster, wave roll-out, hypercare and center of excellence.

12.3 Training and support

Role-based training paths, in-app tutorials, sandbox environment, 24/7 support for Enterprise, status page and SLA reporting.

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13. Acceptance Criteria

1. A new site can be created from a template and take its first order within 1 business day (Starter) / 1 hour after hardware arrives (Growth+ clone).

2. POS continues all core functions with the internet disconnected for 72 hours and syncs without duplicate or lost transactions.
3. A sale correctly updates stock by recipe, posts accounting entries and appears in HQ dashboards within 60 seconds of connectivity.
4. Users can see only the workspaces and data granted to them; cross-tenant access tests all fail.
5. Day-end close produces Z-report, cash reconciliation and ledger postings with zero manual re-keying.
6. Pilot customers achieve measurable improvement: food-cost variance reduced and reporting time cut within 90 days.

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14. Risks and Mitigations

Risk	Mitigation
Poor connectivity at sites	Offline-first design, site hub, lightweight clients
Complexity overwhelming small restaurants	Tiering, feature flags, simplified starter workspace, wizards
Data quality (recipes, units, opening stock)	Import validation, guided counts, templates
Hardware diversity	Certified hardware list, universal device agent
Regulatory differences	Localization packs, configurable tax/fiscal engine, local partners
Staff resistance	Role-based training, simple UI, quick wins (faster service, less paperwork)
Fraud and shrinkage	Audit trail, approval limits, anomaly alerts, blind counts
Scope creep	Modular roadmap, prioritization by tier and customer value

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15. Traceability and Next Steps

- Each requirement ID (POS-xx, INV-xx, FIN-xx, etc.) maps to user stories, test cases and release milestones.
- **Next steps:** (1) validate priorities with 3–5 pilot restaurants (one small, one medium, one chain); (2) produce UX prototypes of the Site Workspace and HQ Workspace; (3) finalize technical architecture and sync design; (4) build the Phase 1 backlog and estimate; (5) select localization targets (tax, fiscal, payments) for the first launch markets.

Appendix A – Example Permission Matrix (excerpt)

Action	Cashier	Waiter	Chef	Storekeeper	Site Mgr	Area Mgr	Finance	Owner
Take order	✔	✔	–	–	✔	–	–	–
Void item	Limit	Limit	–	–	✔	✔	–	✔
Apply discount	Limit	–	–	–	✔	✔	–	✔
Receive goods	–	–	–	✔	✔	–	–	–
Approve PO	–	–	–	–	Limit	✔	✔	✔
Edit recipe	–	–	Propose	–	Propose	✔	–	✔
Close day	–	–	–	–	✔	–	–	–
View all sites	–	–	–	–	–	Region	✔	✔
Run payroll	–	–	–	–	–	–	✔	Approve

Appendix B – Typical Site Day Flow

Open → Checklist → Cash float → Prep list → Service (orders, KDS, deliveries) → Receiving → Waste logging → Close → Cash count → Z-report → Auto-posting to finance → HQ summary.